Guidance and Control Models for the HL-20 Glide Vehicle

Point-mass dynamics, linearization, error-state models for LQR,
and inner-loop attitude control (classical and NDI)

hgv3_gc project notes

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Contents

How to read this document **3**

1 Notation, frames and conventions **4**
1.1 Reference frames . . . 4
1.2 States, inputs and parameters . . . 4
1.3 Constants . . . 5
1.4 Sign conventions . . . 5

2 Nonlinear point-mass equations of motion **6**
2.1 Newton's second law in the inertial frame . . . 6
2.2 Moving to the rotating Earth frame . . . 6
2.3 Spherical coordinates and the local basis . . . 7
2.4 Translational kinematics . . . 7
2.5 Velocity dynamics . . . 8
2.6 Forces . . . 8
2.6.1 Aerodynamic force . . . 8
2.6.2 Gravity with J_2 . . . 9
2.6.3 Coriolis and centripetal terms . . . 9
2.7 The assembled nonlinear model . . . 10
2.8 Aerodynamic and atmospheric sub-models . . . 10
2.9 Relation to the `hgv3` simulator . . . 11

3 Linearization about a planned trajectory **12**
3.1 Reference trajectory and perturbations . . . 12
3.2 Simplifications for the hand derivation . . . 12
3.3 Aerodynamic partial derivatives . . . 13
3.4 Partial derivatives of the kinematic rows . . . 14
3.5 Partial derivatives of the dynamic rows . . . 14
3.6 The linear model . . . 15
3.7 Structure of the linear model . . . 15

4 Error-state model for path following **16**
4.1 Reference path . . . 16
4.2 Error states . . . 17
4.3 Rate of progress along the path . . . 17
4.4 Nonlinear error dynamics . . . 17
4.5 Linearization . . . 18
4.6 Feedforward and why the model has no disturbance term . . . 19

4.7 Arc length as the independent variable . . . 19
4.8 Structure and controllability . . . 20
4.9 What was neglected . . . 20

5 LQR design on the error-state model **21**
5.1 Cost and weights . . . 21
5.2 Computing the gains . . . 21
5.3 The control law . . . 22
5.4 Getting A and B numerically . . . 22
5.5 Checking the design . . . 22

6 Inner-loop attitude control: classical and NDI **23**
6.1 Notation for this chapter . . . 23
6.2 Rotational dynamics . . . 23
6.3 Aerodynamic moments . . . 24
6.4 Wind-axis attitude kinematics . . . 24
6.5 Classical gain-scheduled design . . . 25
6.5.1 Linearization about a wings-level trim . . . 25
6.5.2 Control structure . . . 25
6.6 Nonlinear dynamic inversion (NDI) . . . 26
6.6.1 The idea, written out . . . 26
6.6.2 Slow loop: angles to body rates . . . 26
6.6.3 Fast loop: body rates to control moments . . . 27
6.6.4 Incremental NDI (INDI) . . . 28
6.6.5 Practical points . . . 28
6.7 Comparison . . . 28

A Verification record **29**

How to read this document

This document derives, in long form, the models used by the trajectory planner and the models needed to close the loop around the plan. The planner code is in `waypoint_traj_solver.py` and `hl20_model.py`.

The structure follows the pattern of Rajamani's *Vehicle Dynamics and Control*: one chapter derives the vehicle model from first principles, and a later chapter goes back to the same nonlinear equations, substitutes error states, and works them into a state-space model for LQR.

Chapter 1 Frames, states, constants and sign conventions.

Chapter 2 The nonlinear point-mass equations of motion over a rotating, oblate Earth, derived from Newton's second law. The result is exactly the model in `hl20_model.py`. The chapter ends with how it differs from the `hgv3` simulator.

Chapter 3 Jacobian linearization of those equations about a planned trajectory, every partial derivative written out.

Chapter 4 The error-state (path-following) model: reference indexed by ground-track arc length, error substitution, and linearization into $\dot{\boldsymbol{e}} = A_e \boldsymbol{e} + B_e \, \delta \boldsymbol{u}$.

Chapter 5 Using that model for LQR gain design, and computing the matrices numerically with CasADi.

Chapter 6 The inner (attitude) loop for a 6-DOF simulation: rigid-body rotational dynamics, wind-axis kinematics, classical gain-scheduled design, NDI and INDI, all in long form.

Appendix A What was checked and how.

Equations are written component by component rather than in compact vector form, and intermediate steps are kept. Every equation in Chapters 2–4 and the kinematics in Chapter 6 were checked symbolically or numerically (Appendix A).

Chapter 1

Notation, frames and conventions

1.1 Reference frames

Inertial (ECI), I. Non-rotating, origin at Earth's center. The `hgv3` simulator integrates its equations of motion here.

Earth-fixed (ECEF), E. Rotates with Earth at constant rate Ω about its z axis $\hat{\mathbf{z}}_E$. Positions and velocities in the planner are Earth-relative.

Local geocentric frame, $(\hat{\mathbf{u}}, \hat{\mathbf{e}}, \hat{\mathbf{n}})$. At the vehicle's position: $\hat{\mathbf{u}}$ points radially up, $\hat{\mathbf{e}}$ east, $\hat{\mathbf{n}}$ north (perpendicular to $\hat{\mathbf{u}}$, toward the pole). It is right-handed in the order $(\hat{\mathbf{u}}, \hat{\mathbf{e}}, \hat{\mathbf{n}})$:

$$\hat{\mathbf{u}} \times \hat{\mathbf{e}} = \hat{\mathbf{n}}, \qquad \hat{\mathbf{e}} \times \hat{\mathbf{n}} = \hat{\mathbf{u}}, \qquad \hat{\mathbf{n}} \times \hat{\mathbf{u}} = \hat{\mathbf{e}}.$$

It is *geocentric*: $\hat{\mathbf{u}}$ is along the radius vector, not the ellipsoid normal.

Velocity frame, $(\hat{\mathbf{t}}, \hat{\mathbf{g}}, \hat{\mathbf{h}})$. $\hat{\mathbf{t}}$ along the Earth-relative velocity, $\hat{\mathbf{g}}$ perpendicular to it in the vertical plane on the "up" side, and $\hat{\mathbf{h}}$ horizontal and to the right of the velocity. Defined in Section 2.5.

Body and wind axes. Used only in Chapter 6.

1.2 States, inputs and parameters

Symbol	Meaning	Units	Code name
r	geocentric radius	m	`r`
θ	longitude	rad	`lon`
ϕ	geocentric latitude	rad	`lat`
V	Earth-relative speed	m/s	`V`
γ	flight-path angle, positive climbing	rad	`gam`
ψ	heading, clockwise from north	rad	`psi`
α	angle of attack	rad	`alpha`
σ	bank angle, positive right wing down	rad	`sigma`
$u_\alpha,\ u_\sigma$	$\dot{\alpha},\ \dot{\sigma}$ (planner controls only)	rad/s	`U`
h	geodetic altitude	m	`h`
$a_L,\ a_D$	lift and drag *accelerations* (force per unit mass)	m/s^2	`L`, `D`
$\bar{q}$	dynamic pressure $\frac{1}{2}\rho V^2$	Pa	`qbar`
M	Mach number $V/a_s(h)$	–	`mach`

The planner uses the 8-element state $[r, \theta, \phi, V, \gamma, \psi, \alpha, \sigma]$ with $[u_\alpha, u_\sigma]$ as controls, so that α and σ are smooth and rate-limited. For the guidance problem in Chapters 3 to 5, α and σ are treated directly as the inputs, because the 3-DOF simulator flies them exactly and a 6-DOF autopilot tracks them.

1.3 Constants

Symbol	Value	Meaning	Source (`hgv3.`)
μ	$3.986004418 \times 10^{14}$ m^3/s^2	gravitational parameter	`MU_EARTH`
R_E	6 378 137 m	equatorial radius	`R_EARTH`
J_2	1.08263×10^{-3}	oblateness coefficient	`J2`
Ω	7.2921159×10^{-5} rad/s	Earth rotation rate	`OMEGA_EARTH`
g_0	9.80665 m/s^2	standard gravity	`G0`
a_W	6 378 137 m	WGS84 semi-major axis	`WGS84_A`
f_W	1/298.257223563	WGS84 flattening	`WGS84_F`
m	8664 kg	HL-20 mass	airframe
S	26.61 m^2	reference area	airframe
$\bar{c}$	8.61 m	reference length (pitch)	airframe
b	4.23 m	reference span (roll, yaw)	airframe
I_{xx}, I_{yy}, I_{zz}	10 185, 45 548, 48 327 kg m^2	principal inertias	airframe
I_{xz}	0 (assumed in the data)	product of inertia	airframe

1.4 Sign conventions

- $\gamma > 0$ is climbing; entry trajectories have $\gamma < 0$.
- ψ is measured clockwise from north: $\psi = \pi/2$ is due east.
- $\sigma > 0$ (right wing down) tilts the lift vector to the right of the velocity and turns the vehicle right, so $\dot{\psi}$ increases. This matches `hgv3`.
- Latitude in the planner is *geocentric*. Waypoints and the simulator use *geodetic* latitude; conversions go through ECEF (`hgv3.lla_to_ecef`).

Chapter 2

Nonlinear point-mass equations of motion

The vehicle is modeled as a point mass acted on by aerodynamic force and gravity. There is no thrust. Angle of attack α and bank σ orient the aerodynamic force; their own dynamics are the subject of Chapter 6.

2.1 Newton's second law in the inertial frame

With $\boldsymbol{r}$ the position vector from Earth's center, Newton's law in ECI is

$$\left.\frac{d^2\boldsymbol{r}}{dt^2}\right|_I = \boldsymbol{a}_{\text{aero}} + \boldsymbol{g}(\boldsymbol{r}), \tag{2.1}$$

where $\boldsymbol{a}_{\text{aero}}$ is the aerodynamic force per unit mass and $\boldsymbol{g}$ is the gravitational acceleration. This is the form the `hgv3` simulator integrates, with RK4 on ECI position and velocity.

2.2 Moving to the rotating Earth frame

For any vector $\boldsymbol{b}$, the derivatives seen in the inertial and Earth frames are related by the transport theorem,

$$\left.\frac{d\boldsymbol{b}}{dt}\right|_I = \left.\frac{d\boldsymbol{b}}{dt}\right|_E + \boldsymbol{\Omega}\times\boldsymbol{b}, \qquad \boldsymbol{\Omega} = \Omega\,\hat{\mathbf{z}}_E. \tag{2.2}$$

Applied to $\boldsymbol{r}$, it gives the inertial velocity in terms of the Earth-relative velocity $\boldsymbol{v} \equiv d\boldsymbol{r}/dt|_E$:

$$\left.\frac{d\boldsymbol{r}}{dt}\right|_I = \boldsymbol{v} + \boldsymbol{\Omega}\times\boldsymbol{r}. \tag{2.3}$$

Applying it a second time, with Ω constant:

$$\begin{aligned}
\left.\frac{d^2\boldsymbol{r}}{dt^2}\right|_I &= \left.\frac{d}{dt}\big(\boldsymbol{v} + \boldsymbol{\Omega}\times\boldsymbol{r}\big)\right|_E + \boldsymbol{\Omega}\times\big(\boldsymbol{v} + \boldsymbol{\Omega}\times\boldsymbol{r}\big) \\
&= \left.\frac{d\boldsymbol{v}}{dt}\right|_E + \boldsymbol{\Omega}\times\boldsymbol{v} + \boldsymbol{\Omega}\times\boldsymbol{v} + \boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\boldsymbol{r}) \\
&= \left.\frac{d\boldsymbol{v}}{dt}\right|_E + 2\,\boldsymbol{\Omega}\times\boldsymbol{v} + \boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\boldsymbol{r}).
\end{aligned} \tag{2.4}$$

Substituting into Equation (2.1) and solving for the Earth-relative acceleration:

$$\boxed{\left.\frac{d\boldsymbol{v}}{dt}\right|_E = \boldsymbol{a}_{\text{aero}} + \boldsymbol{g} \underbrace{-2\,\boldsymbol{\Omega}\times\boldsymbol{v}}_{\text{Coriolis}} \underbrace{-\,\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\boldsymbol{r})}_{\text{centripetal}}} \tag{2.5}$$

2.3 Spherical coordinates and the local basis

The position is described by radius r, longitude θ and geocentric latitude ϕ: $\boldsymbol{r} = r\,\hat{\mathbf{u}}$. In ECEF components the local unit vectors are

$$\hat{\mathbf{u}} = \begin{bmatrix}\cos\phi\cos\theta\\ \cos\phi\sin\theta\\ \sin\phi\end{bmatrix}, \qquad \hat{\mathbf{e}} = \begin{bmatrix}-\sin\theta\\ \cos\theta\\ 0\end{bmatrix}, \qquad \hat{\mathbf{n}} = \begin{bmatrix}-\sin\phi\cos\theta\\ -\sin\phi\sin\theta\\ \cos\phi\end{bmatrix}. \tag{2.6}$$

Earth's spin axis, written in the local basis, is

$$\hat{\mathbf{z}}_E = \sin\phi\,\hat{\mathbf{u}} + \cos\phi\,\hat{\mathbf{n}}, \qquad \text{so} \qquad \boldsymbol{\Omega} = \Omega\sin\phi\,\hat{\mathbf{u}} + \Omega\cos\phi\,\hat{\mathbf{n}}. \tag{2.7}$$

As the vehicle moves, θ and ϕ change and the local basis turns. Differentiating the component expressions above with respect to time (in the Earth frame) gives

$$\frac{d\hat{\mathbf{u}}}{dt} = \dot{\phi}\,\hat{\mathbf{n}} + \dot{\theta}\cos\phi\,\hat{\mathbf{e}}, \tag{2.8}$$

$$\frac{d\hat{\mathbf{e}}}{dt} = -\dot{\theta}\cos\phi\,\hat{\mathbf{u}} + \dot{\theta}\sin\phi\,\hat{\mathbf{n}}, \tag{2.9}$$

$$\frac{d\hat{\mathbf{n}}}{dt} = -\dot{\phi}\,\hat{\mathbf{u}} - \dot{\theta}\sin\phi\,\hat{\mathbf{e}}. \tag{2.10}$$

All three are the cross product of one angular velocity with the basis vector, $d\boldsymbol{b}/dt = \boldsymbol{\omega}_{LE}\times\boldsymbol{b}$, where

$$\boldsymbol{\omega}_{LE} = \dot{\theta}\sin\phi\,\hat{\mathbf{u}} \;-\; \dot{\phi}\,\hat{\mathbf{e}} \;+\; \dot{\theta}\cos\phi\,\hat{\mathbf{n}} \tag{2.11}$$

is the angular velocity of the local frame relative to Earth.

2.4 Translational kinematics

Differentiating $\boldsymbol{r} = r\hat{\mathbf{u}}$ in the Earth frame and using Equation (2.8):

$$\boldsymbol{v} = \dot{r}\,\hat{\mathbf{u}} + r\frac{d\hat{\mathbf{u}}}{dt} = \dot{r}\,\hat{\mathbf{u}} + r\dot{\phi}\,\hat{\mathbf{n}} + r\dot{\theta}\cos\phi\,\hat{\mathbf{e}}. \tag{2.12}$$

The speed V, flight-path angle γ and heading ψ are *defined* by writing the same velocity as

$$\boldsymbol{v} = V\hat{\mathbf{t}}, \qquad \hat{\mathbf{t}} = \sin\gamma\,\hat{\mathbf{u}} + \cos\gamma\cos\psi\,\hat{\mathbf{n}} + \cos\gamma\sin\psi\,\hat{\mathbf{e}}. \tag{2.13}$$

Matching the $\hat{\mathbf{u}}$, $\hat{\mathbf{n}}$ and $\hat{\mathbf{e}}$ components of Equations (2.12) and (2.13):

$$\dot{r} = V\sin\gamma, \tag{2.14}$$

$$\dot{\phi} = \frac{V\cos\gamma\cos\psi}{r}, \tag{2.15}$$

$$\dot{\theta} = \frac{V\cos\gamma\sin\psi}{r\cos\phi}. \tag{2.16}$$

2.5 Velocity dynamics

Two more unit vectors complete an orthonormal triad with $\hat{\mathbf{t}}$. They are the directions in which $\hat{\mathbf{t}}$ moves when γ or ψ changes:

$$\hat{\mathbf{g}} \equiv \frac{\partial \hat{\mathbf{t}}}{\partial \gamma} = \cos\gamma\,\hat{\mathbf{u}} - \sin\gamma\cos\psi\,\hat{\mathbf{n}} - \sin\gamma\sin\psi\,\hat{\mathbf{e}}, \tag{2.17}$$

$$\hat{\mathbf{h}} \equiv \frac{1}{\cos\gamma}\frac{\partial \hat{\mathbf{t}}}{\partial \psi} = -\sin\psi\,\hat{\mathbf{n}} + \cos\psi\,\hat{\mathbf{e}}. \tag{2.18}$$

$\hat{\mathbf{g}}$ lies in the vertical plane, perpendicular to the velocity on the upper side; $\hat{\mathbf{h}}$ is horizontal and to the right of the velocity. One can check that $|\hat{\mathbf{g}}| = |\hat{\mathbf{h}}| = 1$ and $\hat{\mathbf{t}}\cdot\hat{\mathbf{g}} = \hat{\mathbf{t}}\cdot\hat{\mathbf{h}} = \hat{\mathbf{g}}\cdot\hat{\mathbf{h}} = 0$.

Differentiate $\boldsymbol{v} = V\hat{\mathbf{t}}$ in the Earth frame. The unit vector $\hat{\mathbf{t}}$ changes for two reasons: its components change because γ and ψ change, and the local basis it is written in turns with $\boldsymbol{\omega}_{LE}$. So

$$\left.\frac{d\hat{\mathbf{t}}}{dt}\right|_E = \dot{\gamma}\,\frac{\partial \hat{\mathbf{t}}}{\partial \gamma} + \dot{\psi}\,\frac{\partial \hat{\mathbf{t}}}{\partial \psi} + \boldsymbol{\omega}_{LE}\times\hat{\mathbf{t}} = \dot{\gamma}\,\hat{\mathbf{g}} + \dot{\psi}\cos\gamma\,\hat{\mathbf{h}} + \boldsymbol{\omega}_{LE}\times\hat{\mathbf{t}}, \tag{2.19}$$

$$\left.\frac{d\boldsymbol{v}}{dt}\right|_E = \dot{V}\,\hat{\mathbf{t}} + V\dot{\gamma}\,\hat{\mathbf{g}} + V\dot{\psi}\cos\gamma\,\hat{\mathbf{h}} + V\,\boldsymbol{\omega}_{LE}\times\hat{\mathbf{t}}. \tag{2.20}$$

Transport terms. Substituting Equations (2.15) and (2.16) into Equation (2.11) and projecting $\boldsymbol{\omega}_{LE}\times\hat{\mathbf{t}}$ onto the triad gives

$$\hat{\mathbf{t}}\cdot(\boldsymbol{\omega}_{LE}\times\hat{\mathbf{t}}) = 0, \tag{2.21}$$

$$\hat{\mathbf{g}}\cdot(\boldsymbol{\omega}_{LE}\times\hat{\mathbf{t}}) = -\frac{V}{r}\cos\gamma, \tag{2.22}$$

$$\hat{\mathbf{h}}\cdot(\boldsymbol{\omega}_{LE}\times\hat{\mathbf{t}}) = -\frac{V}{r}\cos^2\gamma\,\sin\psi\tan\phi. \tag{2.23}$$

These are the effects of flying over a curved Earth: the local horizon tilts as the vehicle moves, and the direction of north changes with longitude.

Projection. Let $\boldsymbol{a}$ be the right-hand side of Equation (2.5). Dotting Equation (2.20) with $\hat{\mathbf{t}}$, $\hat{\mathbf{g}}$ and $\hat{\mathbf{h}}$ in turn, and moving the transport terms to the right:

$$\dot{V} = \hat{\mathbf{t}}\cdot\boldsymbol{a}, \tag{2.24}$$

$$\dot{\gamma} = \frac{1}{V}\,\hat{\mathbf{g}}\cdot\boldsymbol{a} + \frac{V}{r}\cos\gamma, \tag{2.25}$$

$$\dot{\psi} = \frac{1}{V\cos\gamma}\,\hat{\mathbf{h}}\cdot\boldsymbol{a} + \frac{V}{r}\cos\gamma\sin\psi\tan\phi. \tag{2.26}$$

What remains is to project each force term of $\boldsymbol{a}$.

2.6 Forces

2.6.1 Aerodynamic force

Drag acts opposite the velocity. Lift is perpendicular to it, rotated from $\hat{\mathbf{g}}$ toward $\hat{\mathbf{h}}$ by the bank angle:

$$\boldsymbol{a}_{\text{aero}} = -a_D\,\hat{\mathbf{t}} + a_L\big(\cos\sigma\,\hat{\mathbf{g}} + \sin\sigma\,\hat{\mathbf{h}}\big), \tag{2.27}$$

$$a_L = \frac{\rho(h)\,V^2 S}{2m}\,C_L(\alpha, M), \qquad a_D = \frac{\rho(h)\,V^2 S}{2m}\,C_D(\alpha, M), \qquad M = \frac{V}{a_s(h)}. \tag{2.28}$$

Projections:

$$\hat{\mathbf{t}} \cdot \boldsymbol{a}_{\text{aero}} = -a_D, \qquad \hat{\mathbf{g}} \cdot \boldsymbol{a}_{\text{aero}} = a_L \cos\sigma, \qquad \hat{\mathbf{h}} \cdot \boldsymbol{a}_{\text{aero}} = a_L \sin\sigma.$$

With $\sigma > 0$ the lift has a component along $+\hat{\mathbf{h}}$ (to the right), which turns the vehicle right.

2.6.2 Gravity with J_2

The gravitational potential truncated after J_2 is

$$U(r, \phi) = \frac{\mu}{r}\left[1 - J_2\left(\frac{R_E}{r}\right)^2 P_2(\sin\phi)\right], \qquad P_2(x) = \tfrac{1}{2}\left(3x^2 - 1\right), \tag{2.29}$$

and $\boldsymbol{g} = \nabla U$. In the local frame, the radial and northward components are

$$\hat{\mathbf{u}} \cdot \boldsymbol{g} = \frac{\partial U}{\partial r} = -\frac{\mu}{r^2} + \frac{3\mu J_2 R_E^2}{r^4} P_2(\sin\phi) = -\frac{\mu}{r^2}\left[1 - \frac{3}{2} J_2\left(\frac{R_E}{r}\right)^2 \left(3\sin^2\phi - 1\right)\right], \tag{2.30}$$

$$\hat{\mathbf{n}} \cdot \boldsymbol{g} = \frac{1}{r}\frac{\partial U}{\partial \phi} = \frac{\mu}{r^2}\left(-J_2 \frac{R_E^2}{r^2}\right)\frac{dP_2}{d\phi} = -\frac{3\mu J_2 R_E^2}{r^4}\sin\phi\cos\phi. \tag{2.31}$$

Define the downward radial magnitude g_r and the signed northward component g_n:

$$g_r = \frac{\mu}{r^2}\left[1 - \frac{3}{2} J_2\left(\frac{R_E}{r}\right)^2 \left(3\sin^2\phi - 1\right)\right], \qquad g_n = -\frac{3\mu J_2 R_E^2}{r^4}\sin\phi\cos\phi, \qquad \boldsymbol{g} = -g_r\,\hat{\mathbf{u}} + g_n\,\hat{\mathbf{n}}. \tag{2.32}$$

Projections:

$$\hat{\mathbf{t}} \cdot \boldsymbol{g} = -g_r \sin\gamma + g_n \cos\gamma\cos\psi, \qquad \hat{\mathbf{g}} \cdot \boldsymbol{g} = -g_r\cos\gamma - g_n\sin\gamma\cos\psi, \qquad \hat{\mathbf{h}} \cdot \boldsymbol{g} = -g_n\sin\psi.$$

2.6.3 Coriolis and centripetal terms

Using Equation (2.7) for $\boldsymbol{\Omega}$, $\boldsymbol{v} = V\hat{\mathbf{t}}$ and $\boldsymbol{r} = r\hat{\mathbf{u}}$, the Coriolis acceleration $-2\boldsymbol{\Omega} \times \boldsymbol{v}$ projects as

$$\hat{\mathbf{t}} \cdot (-2\boldsymbol{\Omega} \times \boldsymbol{v}) = 0, \tag{2.33}$$

$$\hat{\mathbf{g}} \cdot (-2\boldsymbol{\Omega} \times \boldsymbol{v}) = 2\Omega V \cos\phi \sin\psi, \tag{2.34}$$

$$\hat{\mathbf{h}} \cdot (-2\boldsymbol{\Omega} \times \boldsymbol{v}) = 2\Omega V\left(\sin\phi\cos\gamma - \cos\phi\sin\gamma\cos\psi\right), \tag{2.35}$$

and the centripetal acceleration $-\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \boldsymbol{r})$ as

$$\hat{\mathbf{t}} \cdot (-\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \boldsymbol{r})) = \Omega^2 r\cos\phi\left(\sin\gamma\cos\phi - \cos\gamma\sin\phi\cos\psi\right), \tag{2.36}$$

$$\hat{\mathbf{g}} \cdot (-\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \boldsymbol{r})) = \Omega^2 r\cos\phi\left(\cos\gamma\cos\phi + \sin\gamma\sin\phi\cos\psi\right), \tag{2.37}$$

$$\hat{\mathbf{h}} \cdot (-\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \boldsymbol{r})) = \Omega^2 r\sin\phi\cos\phi\sin\psi. \tag{2.38}$$

The Coriolis term has no component along the velocity, so it never changes the speed.

2.7 The assembled nonlinear model

Substituting all projections into Equations (2.24) to (2.26) and collecting the kinematics Equations (2.14) to (2.16):

$$\dot{r} = V \sin\gamma \tag{2.39}$$

$$\dot{\theta} = \frac{V \cos\gamma \sin\psi}{r \cos\phi} \tag{2.40}$$

$$\dot{\phi} = \frac{V \cos\gamma \cos\psi}{r} \tag{2.41}$$

$$\dot{V} = -a_D - g_r \sin\gamma + g_n \cos\gamma \cos\psi + \Omega^2 r \cos\phi \,(\sin\gamma \cos\phi - \cos\gamma \sin\phi \cos\psi) \tag{2.42}$$

$$\begin{aligned}\dot{\gamma} = {} & \frac{a_L \cos\sigma}{V} - \frac{g_r \cos\gamma}{V} - \frac{g_n \sin\gamma \cos\psi}{V} + \frac{V \cos\gamma}{r} + 2\Omega \cos\phi \sin\psi \\ & + \frac{\Omega^2 r \cos\phi}{V} \,(\cos\gamma \cos\phi + \sin\gamma \sin\phi \cos\psi)\end{aligned} \tag{2.43}$$

$$\begin{aligned}\dot{\psi} = {} & \frac{a_L \sin\sigma}{V \cos\gamma} - \frac{g_n \sin\psi}{V \cos\gamma} + \frac{V \cos\gamma \sin\psi \tan\phi}{r} - 2\Omega \,(\cos\phi \cos\psi \tan\gamma - \sin\phi) \\ & + \frac{\Omega^2 r \sin\phi \cos\phi \sin\psi}{V \cos\gamma}\end{aligned} \tag{2.44}$$

In the planner, two more equations make α and σ states:

$$\dot{\alpha} = u_\alpha, \qquad \dot{\sigma} = u_\sigma. \tag{2.45}$$

These are line for line the expressions in `HL20Model.xdot`.

2.8 Aerodynamic and atmospheric sub-models

- **Altitude.** The atmosphere is a function of geodetic altitude. The model uses the height above the WGS84 ellipsoid along the radius,

$$h = r - R_{\text{ell}}(\phi), \qquad R_{\text{ell}}(\phi) = \frac{a_W b_W}{\sqrt{b_W^2 \cos^2\phi + a_W^2 \sin^2\phi}}, \qquad b_W = a_W(1 - f_W), \tag{2.46}$$

  which differs from the true geodetic altitude by much less than a meter here. Using $r - R_E$ instead would be wrong by about 5 km at 30° latitude, which is a factor of about 2 in density.

- **Atmosphere.** $\ln\rho(h)$ and $a_s(h)$ are B-spline fits of `hgv3.atmosphere` (USSA76) sampled every 500 m (`atmosphereFits`).

- **Aerodynamics.** $C_L(\alpha, M)$ and $C_D(\alpha, M)$ are 2-D B-spline fits of `hgv3.wind_coefficients` on an (α, M) grid (`aeroFits`). The splines give the continuous first and second derivatives the NLP solver needs; the simulator's own table is linear in Mach with clamped α.

- **Valid α window.** `hgv3` clamps α to each Mach row's range and blends rows linearly in Mach. The window in which no clamping occurs is the intersection of the two bracketing rows' ranges; `alphaLimitFits` builds a continuous, conservative version $\alpha_{\text{lo}}(M) \le \alpha \le \alpha_{\text{hi}}(M)$.

2.9 Relation to the `hgv3` simulator

`hgv3` integrates Equation (2.1) in ECI with fixed-step RK4. The physics is the same (same aero table, USSA76, point-mass plus J_2 gravity), with two implementation differences that matter:

1. It co-rotates the atmosphere about the ECI z axis, while its ECEF frame comes from the full ERFA precession–nutation chain. In 2026 the two axes differ by 0.148°, so the simulated air mass moves about 1.2 m/s relative to ECEF at the equator. The model above co-rotates about the ECEF pole.

2. Its aero coefficients are piecewise linear in Mach with α clamping, where the model uses smooth fits.

Flying the same open-loop (α, σ) for 800 s from Mach 15, the two agree to about 10 m in altitude and 0.1 m/s in speed, with an along-track drift of about 0.8 m/s (670 m after 800 s) consistent with item 1.

Chapter 3

Linearization about a planned trajectory

3.1 Reference trajectory and perturbations

For guidance, write the model as $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u})$ with

$$\boldsymbol{x} = [\, r,\ \theta,\ \phi,\ V,\ \gamma,\ \psi \,]^T, \qquad \boldsymbol{u} = [\, \alpha,\ \sigma \,]^T.$$

The planner supplies a reference $\boldsymbol{x}^*(t)$, $\boldsymbol{u}^*(t)$ that satisfies the model: $\dot{\boldsymbol{x}}^* = \boldsymbol{f}(\boldsymbol{x}^*, \boldsymbol{u}^*)$. Define the perturbations

$$\delta\boldsymbol{x} = \boldsymbol{x} - \boldsymbol{x}^*(t), \qquad \delta\boldsymbol{u} = \boldsymbol{u} - \boldsymbol{u}^*(t).$$

A first-order Taylor expansion of $\boldsymbol{f}$ about the reference gives

$$\delta\dot{\boldsymbol{x}} = A(t)\,\delta\boldsymbol{x} + B(t)\,\delta\boldsymbol{u}, \qquad A(t) = \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right|_*, \qquad B(t) = \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{u}}\right|_*, \tag{3.1}$$

where $|_*$ means evaluated on the reference. The zeroth-order terms cancel precisely because the reference satisfies the model.

3.2 Simplifications for the hand derivation

For the hand derivation only, Earth rotation and J_2 are dropped, and altitude is taken as $h = r - R_0$ with R_0 the local ellipsoid radius held constant:

$$\Omega = 0, \qquad J_2 = 0 \ \Rightarrow\ g_r = g \equiv \frac{\mu}{r^2},\ g_n = 0, \qquad \frac{\partial h}{\partial r} = 1,\ \frac{\partial h}{\partial \phi} = 0. \tag{3.2}$$

This was checked on the test plan by computing exact Jacobians of the full model and of the simplified one with CasADi:

- With states scaled to typical error sizes (1 km of position, 10 m/s, 1°), the largest entry of A is about 0.11 s^{-1}, and rotation plus J_2 change any entry by at most 5×10^{-5} s^{-1}. B is unchanged.
- The open-loop eigenvalues move by at most 2×10^{-3} s^{-1}.

- The neglected latitude dependence of h is about 3 m of altitude per km of northward displacement.

Rotation and J_2 do dominate a few entries that are themselves tiny (for example $\partial\dot{\gamma}/\partial\psi$ comes only from the Coriolis term), but those entries correspond to time scales of hours. The simplified model therefore captures the structure that matters for control. In code, use the exact Jacobians (Section 5.4).

The simplified equations are:

$$\dot{r} = V\sin\gamma \tag{3.3}$$

$$\dot{\theta} = \frac{V\cos\gamma\sin\psi}{r\cos\phi} \tag{3.4}$$

$$\dot{\phi} = \frac{V\cos\gamma\cos\psi}{r} \tag{3.5}$$

$$\dot{V} = -a_D - g\sin\gamma \tag{3.6}$$

$$\dot{\gamma} = \frac{a_L\cos\sigma}{V} - \frac{g\cos\gamma}{V} + \frac{V\cos\gamma}{r} \tag{3.7}$$

$$\dot{\psi} = \frac{a_L\sin\sigma}{V\cos\gamma} + \frac{V\cos\gamma\sin\psi\tan\phi}{r} \tag{3.8}$$

3.3 Aerodynamic partial derivatives

Write $k \equiv \bar{q}S/m = \rho V^2 S/(2m)$, so that $a_L = k\,C_L(\alpha, M)$ and $a_D = k\,C_D(\alpha, M)$. The coefficient slopes and atmospheric gradients are

$$C_{L,\alpha} = \frac{\partial C_L}{\partial\alpha}, \quad C_{L,M} = \frac{\partial C_L}{\partial M}, \quad C_{D,\alpha} = \frac{\partial C_D}{\partial\alpha}, \quad C_{D,M} = \frac{\partial C_D}{\partial M},$$

$$H_\rho \equiv -\frac{\rho}{d\rho/dh} \quad \text{(local density scale height)}, \qquad a_s' \equiv \frac{da_s}{dh}.$$

Partials of the intermediate quantities:

$$\frac{\partial k}{\partial V} = \frac{\rho V S}{m} = \frac{2k}{V}, \qquad \frac{\partial k}{\partial h} = \frac{d\rho}{dh}\frac{V^2 S}{2m} = -\frac{k}{H_\rho}, \tag{3.9}$$

$$\frac{\partial M}{\partial V} = \frac{1}{a_s}, \qquad \frac{\partial M}{\partial h} = -\frac{V a_s'}{a_s^2}. \tag{3.10}$$

By the chain rule, with $a_L = k\,C_L(\alpha, M(V,h))$:

$$\frac{\partial a_L}{\partial V} = \frac{\partial k}{\partial V}C_L + k\,C_{L,M}\frac{\partial M}{\partial V} = \frac{2a_L}{V} + \frac{k\,C_{L,M}}{a_s}, \tag{3.11}$$

$$\frac{\partial a_L}{\partial h} = \frac{\partial k}{\partial h}C_L + k\,C_{L,M}\frac{\partial M}{\partial h} = -\frac{a_L}{H_\rho} - \frac{k\,C_{L,M}\,V a_s'}{a_s^2}, \tag{3.12}$$

$$\frac{\partial a_L}{\partial\alpha} = k\,C_{L,\alpha}, \tag{3.13}$$

and identically for drag:

$$\frac{\partial a_D}{\partial V} = \frac{2a_D}{V} + \frac{k\,C_{D,M}}{a_s}, \qquad \frac{\partial a_D}{\partial h} = -\frac{a_D}{H_\rho} - \frac{k\,C_{D,M}\,V a_s'}{a_s^2}, \qquad \frac{\partial a_D}{\partial\alpha} = k\,C_{D,\alpha}. \tag{3.14}$$

Gravity: $\partial g/\partial r = -2g/r$. Since $h = r - R_0$, a partial with respect to r equals the partial with respect to h.

3.4 Partial derivatives of the kinematic rows

$\dot{r} = V\sin\gamma$**:**

$$\frac{\partial \dot{r}}{\partial V} = \sin\gamma, \qquad \frac{\partial \dot{r}}{\partial \gamma} = V\cos\gamma.$$

$\dot{\theta} = V\cos\gamma\sin\psi/(r\cos\phi)$**:**

$$\frac{\partial \dot{\theta}}{\partial r} = -\frac{V\cos\gamma\sin\psi}{r^2\cos\phi}, \qquad \frac{\partial \dot{\theta}}{\partial \phi} = \frac{V\cos\gamma\sin\psi\tan\phi}{r\cos\phi}, \qquad \frac{\partial \dot{\theta}}{\partial V} = \frac{\cos\gamma\sin\psi}{r\cos\phi},$$
$$\frac{\partial \dot{\theta}}{\partial \gamma} = -\frac{V\sin\gamma\sin\psi}{r\cos\phi}, \qquad \frac{\partial \dot{\theta}}{\partial \psi} = \frac{V\cos\gamma\cos\psi}{r\cos\phi}.$$

$\dot{\phi} = V\cos\gamma\cos\psi/r$**:**

$$\frac{\partial \dot{\phi}}{\partial r} = -\frac{V\cos\gamma\cos\psi}{r^2}, \quad \frac{\partial \dot{\phi}}{\partial V} = \frac{\cos\gamma\cos\psi}{r}, \quad \frac{\partial \dot{\phi}}{\partial \gamma} = -\frac{V\sin\gamma\cos\psi}{r}, \quad \frac{\partial \dot{\phi}}{\partial \psi} = -\frac{V\cos\gamma\sin\psi}{r}.$$

No kinematic row depends on θ, α or σ.

3.5 Partial derivatives of the dynamic rows

Each partial is named a_{ij} (state) or b_{ij} (input) for use in Chapter 4.

Speed, $\dot{V} = -a_D - g\sin\gamma$**:**

$$a_{Vr} = \frac{\partial \dot{V}}{\partial r} = -\frac{\partial a_D}{\partial h} + \frac{2g}{r}\sin\gamma, \qquad a_{VV} = \frac{\partial \dot{V}}{\partial V} = -\frac{\partial a_D}{\partial V}, \qquad a_{V\gamma} = \frac{\partial \dot{V}}{\partial \gamma} = -g\cos\gamma, \tag{3.15}$$
$$b_{V\alpha} = \frac{\partial \dot{V}}{\partial \alpha} = -\frac{\partial a_D}{\partial \alpha}, \qquad b_{V\sigma} = \frac{\partial \dot{V}}{\partial \sigma} = 0.$$

Flight-path angle, $\dot{\gamma} = a_L\cos\sigma/V - g\cos\gamma/V + V\cos\gamma/r$**:**

$$a_{\gamma r} = \frac{\partial \dot{\gamma}}{\partial r} = \frac{\cos\sigma}{V}\frac{\partial a_L}{\partial h} + \left(\frac{2g}{rV} - \frac{V}{r^2}\right)\cos\gamma, \tag{3.16}$$

$$a_{\gamma V} = \frac{\partial \dot{\gamma}}{\partial V} = \frac{\cos\sigma}{V}\frac{\partial a_L}{\partial V} - \frac{a_L\cos\sigma}{V^2} + \left(\frac{g}{V^2} + \frac{1}{r}\right)\cos\gamma, \tag{3.17}$$

$$a_{\gamma\gamma} = \frac{\partial \dot{\gamma}}{\partial \gamma} = \left(\frac{g}{V} - \frac{V}{r}\right)\sin\gamma, \tag{3.18}$$

$$b_{\gamma\alpha} = \frac{\partial \dot{\gamma}}{\partial \alpha} = \frac{\cos\sigma}{V}\frac{\partial a_L}{\partial \alpha}, \qquad b_{\gamma\sigma} = \frac{\partial \dot{\gamma}}{\partial \sigma} = -\frac{a_L\sin\sigma}{V}. \tag{3.19}$$

Heading, $\dot{\psi} = a_L\sin\sigma/(V\cos\gamma) + V\cos\gamma\sin\psi\tan\phi/r$**:**

$$a_{\psi r} = \frac{\partial \dot{\psi}}{\partial r} = \frac{\sin\sigma}{V\cos\gamma}\frac{\partial a_L}{\partial h} - \frac{V}{r^2}\cos\gamma\sin\psi\tan\phi, \tag{3.20}$$

$$a_{\psi\phi} = \frac{\partial \dot{\psi}}{\partial \phi} = \frac{V}{r}\frac{\cos\gamma\sin\psi}{\cos^2\phi}, \tag{3.21}$$

$$a_{\psi V} = \frac{\partial \dot{\psi}}{\partial V} = \frac{\sin\sigma}{V\cos\gamma}\frac{\partial a_L}{\partial V} - \frac{a_L \sin\sigma}{V^2\cos\gamma} + \frac{1}{r}\cos\gamma\sin\psi\tan\phi, \tag{3.22}$$

$$a_{\psi\gamma} = \frac{\partial \dot{\psi}}{\partial \gamma} = \frac{a_L \sin\sigma\sin\gamma}{V\cos^2\gamma} - \frac{V}{r}\sin\gamma\sin\psi\tan\phi, \tag{3.23}$$

$$a_{\psi\psi} = \frac{\partial \dot{\psi}}{\partial \psi} = \frac{V}{r}\cos\gamma\cos\psi\tan\phi, \tag{3.24}$$

$$b_{\psi\alpha} = \frac{\partial \dot{\psi}}{\partial \alpha} = \frac{\sin\sigma}{V\cos\gamma}\frac{\partial a_L}{\partial \alpha}, \qquad b_{\psi\sigma} = \frac{\partial \dot{\psi}}{\partial \sigma} = \frac{a_L\cos\sigma}{V\cos\gamma}. \tag{3.25}$$

The speed and flight-path-angle rows do not depend on θ, ϕ or ψ in the simplified model.

3.6 The linear model

With the state order $[r, \theta, \phi, V, \gamma, \psi]$ and input order $[\alpha, \sigma]$, writing the kinematic partials as $a_{rV} = \partial\dot{r}/\partial V$, and so on:

$$\begin{bmatrix} \delta\dot{r} \\ \delta\dot{\theta} \\ \delta\dot{\phi} \\ \delta\dot{V} \\ \delta\dot{\gamma} \\ \delta\dot{\psi} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & a_{rV} & a_{r\gamma} & 0 \\ a_{\theta r} & 0 & a_{\theta\phi} & a_{\theta V} & a_{\theta\gamma} & a_{\theta\psi} \\ a_{\phi r} & 0 & 0 & a_{\phi V} & a_{\phi\gamma} & a_{\phi\psi} \\ a_{Vr} & 0 & 0 & a_{VV} & a_{V\gamma} & 0 \\ a_{\gamma r} & 0 & 0 & a_{\gamma V} & a_{\gamma\gamma} & 0 \\ a_{\psi r} & 0 & a_{\psi\phi} & a_{\psi V} & a_{\psi\gamma} & a_{\psi\psi} \end{bmatrix} \begin{bmatrix} \delta r \\ \delta\theta \\ \delta\phi \\ \delta V \\ \delta\gamma \\ \delta\psi \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ b_{V\alpha} & 0 \\ b_{\gamma\alpha} & b_{\gamma\sigma} \\ b_{\psi\alpha} & b_{\psi\sigma} \end{bmatrix} \begin{bmatrix} \delta\alpha \\ \delta\sigma \end{bmatrix}. \tag{3.26}$$

3.7 Structure of the linear model

- **Longitude is ignorable.** The θ column is zero: nothing depends on longitude, so it is a pure integrator (an eigenvalue at zero). This is why tracking in latitude and longitude is awkward, and why Chapter 4 uses errors measured along and across the path instead.
- **Longitudinal block.** $(\delta r, \delta V, \delta\gamma)$ form a closed subsystem driven by $\delta\alpha$ (through $b_{V\alpha}$ and $b_{\gamma\alpha}$) and by $\delta\sigma$ (through $b_{\gamma\sigma}$).
- **Lateral block.** $(\delta\theta, \delta\phi, \delta\psi)$ are driven by the longitudinal states and by $\delta\sigma$ through $b_{\psi\sigma}$.
- **Coupling through bank.** When $\sigma^* \approx 0$: $b_{\gamma\sigma} \approx 0$ and $b_{\psi\alpha} \approx 0$, so α controls the longitudinal motion and σ the lateral motion, separately. This is the analog of Rajamani's decoupled lateral and longitudinal vehicle models. With large σ^*, a change in bank shifts lift between vertical and horizontal, which couples the two. Entry guidance exploits exactly this: bank *magnitude* controls range, and bank *sign* controls crossrange.
- **Open-loop modes.** At one point on the test plan (Mach about 11), the eigenvalues are a lightly damped oscillation in speed, altitude and flight-path angle at about 0.026 rad/s (period near 4 min, damping ratio about 0.06; the entry analog of the phugoid), slow modes near 10^{-3} s^{-1} (one slightly unstable), and the zero eigenvalue from longitude.

Chapter 4

Error-state model for path following

This chapter goes back to the nonlinear equations of Section 3.2, substitutes errors measured relative to the reference *path*, and linearizes. It mirrors the derivation of Rajamani's error-state lateral model (e_1, e_2), with the speed loop folded into the same model because a glider's only longitudinal lever is drag.

4.1 Reference path

From the planned trajectory, compute the ground-track arc length s along a sphere of radius R_0. Along the reference,

$$\dot{s}^* = \frac{R_0}{r^*}\, V^* \cos\gamma^*, \tag{4.1}$$

and every reference quantity can be expressed as a function of s: $h^*(s)$, $V^*(s)$, $\gamma^*(s)$, $\psi^*(s)$, $\alpha^*(s)$, $\sigma^*(s)$. Define also the reference rates, which come straight from the model evaluated on the reference,

$$\dot{V}^* = f_V(\boldsymbol{x}^*, \boldsymbol{u}^*), \qquad \dot{\gamma}^* = f_\gamma(\boldsymbol{x}^*, \boldsymbol{u}^*), \qquad \dot{\psi}^* = f_\psi(\boldsymbol{x}^*, \boldsymbol{u}^*),$$

and the geodesic curvature of the ground track, positive for a right turn:

$$\kappa(s) = \frac{d\psi^*}{ds} - \frac{\sin\psi^* \tan\phi^*}{R_0}. \tag{4.2}$$

The second term removes the change in heading caused only by the convergence of meridians; κ is the curvature a local flat map would show.

Reference point. At each instant, $s(t)$ is chosen as the point on the reference ground track closest to the vehicle, so the along-track position error is zero by construction. How that point is found (projection, spline search, integrating $\dot{s}$) is independent of what follows.

4.2 Error states

Error	Definition	Meaning
e_c	signed distance from the reference ground track	crossrange error, positive to the right
e_h	$h - h^*(s)$	altitude error
e_ψ	$\psi - \psi^*(s)$	heading error
e_γ	$\gamma - \gamma^*(s)$	flight-path angle error
e_V	$V - V^*(s)$	speed error at the same path position
$\delta\alpha$	$\alpha - \alpha^*(s)$	angle-of-attack correction
$\delta\sigma$	$\sigma - \sigma^*(s)$	bank correction

The vehicle state is therefore $V = V^* + e_V$, $\gamma = \gamma^* + e_\gamma$, $\psi = \psi^* + e_\psi$, $r = r^* + e_h$, $\alpha = \alpha^* + \delta\alpha$, $\sigma = \sigma^* + \delta\sigma$, all reference values taken at the current s.

4.3 Rate of progress along the path

The vehicle's horizontal velocity, scaled down to the reference sphere, has magnitude $(R_0/r)V\cos\gamma$. Its component along the path direction is that times $\cos e_\psi$. A curve offset by e_c toward the center of a turn of curvature κ is shorter by the factor $(1 - \kappa e_c)$, so the projected point advances faster by the inverse factor:

$$\dot{s} = \frac{R_0}{r}\,\frac{V\cos\gamma\cos e_\psi}{1 - \kappa e_c}. \tag{4.3}$$

The component across the path changes the crossrange error:

$$\dot{e}_c = \frac{R_0}{r}\,V\cos\gamma\sin e_\psi. \tag{4.4}$$

4.4 Nonlinear error dynamics

Differentiate each error. A reference quantity $X^*(s)$ changes only because s changes: $\frac{d}{dt}X^*(s) = \frac{dX^*}{ds}\,\dot{s}$. So

$$\dot{e}_c = \frac{R_0}{r}\,V\cos\gamma\sin e_\psi, \tag{4.5}$$

$$\dot{e}_h = V\sin\gamma - \frac{dh^*}{ds}\,\dot{s}, \tag{4.6}$$

$$\dot{e}_\psi = f_\psi(\boldsymbol{x},\boldsymbol{u}) - \frac{d\psi^*}{ds}\,\dot{s}, \tag{4.7}$$

$$\dot{e}_\gamma = f_\gamma(\boldsymbol{x},\boldsymbol{u}) - \frac{d\gamma^*}{ds}\,\dot{s}, \tag{4.8}$$

$$\dot{e}_V = f_V(\boldsymbol{x},\boldsymbol{u}) - \frac{dV^*}{ds}\,\dot{s}, \tag{4.9}$$

with $\dot{s}$ from Equation (4.3) and f_V, f_γ, f_ψ the right-hand sides of Equations (3.6) to (3.8) evaluated at the *actual* state and input. The reference slopes follow from dividing reference rates by $\dot{s}^*$:

$$\frac{dh^*}{ds} = \frac{V^*\sin\gamma^*}{\dot{s}^*} = \frac{r^*}{R_0}\tan\gamma^*, \qquad \frac{d\psi^*}{ds} = \frac{\dot{\psi}^*}{\dot{s}^*}, \qquad \frac{d\gamma^*}{ds} = \frac{\dot{\gamma}^*}{\dot{s}^*}, \qquad \frac{dV^*}{ds} = \frac{\dot{V}^*}{\dot{s}^*}. \tag{4.10}$$

These play the role of Rajamani's $\dot{\psi}_{\text{des}}$ terms: they are the rates at which the reference itself moves.

4.5 Linearization

Step 1: linearize the path rate

Expand Equation (4.3) to first order about zero error. Each partial is evaluated at the reference:

$$\frac{\partial \dot{s}}{\partial e_V} = \frac{R_0}{r^*}\cos\gamma^* = \frac{\dot{s}^*}{V^*}, \qquad \frac{\partial \dot{s}}{\partial e_\gamma} = -\frac{R_0}{r^*}V^*\sin\gamma^* = -\dot{s}^*\tan\gamma^*,$$
$$\frac{\partial \dot{s}}{\partial e_h} = -\frac{R_0}{r^{*2}}V^*\cos\gamma^* = -\frac{\dot{s}^*}{r^*}, \qquad \frac{\partial \dot{s}}{\partial e_c} = \dot{s}^*\kappa, \qquad \frac{\partial \dot{s}}{\partial e_\psi} = 0.$$

The heading-error partial vanishes because $\partial \cos e_\psi / \partial e_\psi = 0$ at $e_\psi = 0$. So

$$\dot{s} \approx \dot{s}^*\left(1 + \frac{e_V}{V^*} - \tan\gamma^*\, e_\gamma - \frac{e_h}{r^*} + \kappa\, e_c\right). \tag{4.11}$$

Step 2: crossrange row

From Equation (4.5), with $\sin e_\psi \approx e_\psi$:

$$\dot{e}_c = \dot{s}^*\, e_\psi. \tag{4.12}$$

Step 3: altitude row

Linearize Equation (4.6) term by term. The first term gives $\delta(V\sin\gamma) = \sin\gamma^*\, e_V + V^*\cos\gamma^*\, e_\gamma$. The second term is $\frac{r^*}{R_0}\tan\gamma^*$ times Equation (4.11). Collecting by error:

$$\begin{aligned}
e_V: &\quad \sin\gamma^* - \frac{r^*}{R_0}\tan\gamma^*\,\frac{\dot{s}^*}{V^*} = \sin\gamma^* - \sin\gamma^* = 0,\\
e_\gamma: &\quad V^*\cos\gamma^* + \frac{r^*}{R_0}\tan\gamma^*\,\dot{s}^*\tan\gamma^* = V^*\cos\gamma^* + V^*\frac{\sin^2\gamma^*}{\cos\gamma^*} = \frac{V^*}{\cos\gamma^*},\\
e_h: &\quad \frac{r^*}{R_0}\tan\gamma^*\,\frac{\dot{s}^*}{r^*} = \frac{V^*\sin\gamma^*}{r^*},\\
e_c: &\quad -\frac{r^*}{R_0}\tan\gamma^*\,\dot{s}^*\kappa = -V^*\kappa\sin\gamma^*.
\end{aligned}$$

Speed error drops out: going faster along the same path changes where you are along it, not your height above it. So

$$\dot{e}_h = -V^*\kappa\sin\gamma^*\, e_c + \frac{V^*\sin\gamma^*}{r^*}\, e_h + \frac{V^*}{\cos\gamma^*}\, e_\gamma. \tag{4.13}$$

Step 4: heading, flight-path angle and speed rows

For each of the three dynamic rows, $f_\square(\boldsymbol{x}, \boldsymbol{u})$ linearizes to the Chapter 3 partials, and the path-rate term contributes $-\dot{\square}^*/\dot{s}^*$ times Equation (4.11). For heading:

$$\begin{aligned}
\dot{e}_\psi = {} & a_{\psi r}\, e_h + a_{\psi V}\, e_V + a_{\psi\gamma}\, e_\gamma + a_{\psi\psi}\, e_\psi + b_{\psi\alpha}\,\delta\alpha + b_{\psi\sigma}\,\delta\sigma\\
& - \dot{\psi}^*\left(\frac{e_V}{V^*} - \tan\gamma^*\, e_\gamma - \frac{e_h}{r^*} + \kappa\, e_c\right),
\end{aligned} \tag{4.14}$$

and collecting terms:

$$\dot{e}_\psi = -\kappa\,\dot{\psi}^*\, e_c + \left(a_{\psi r} + \frac{\dot{\psi}^*}{r^*}\right) e_h + a_{\psi\psi}\, e_\psi + \left(a_{\psi\gamma} + \dot{\psi}^*\tan\gamma^*\right) e_\gamma + \left(a_{\psi V} - \frac{\dot{\psi}^*}{V^*}\right) e_V$$

$$+ b_{\psi\alpha}\,\delta\alpha + b_{\psi\sigma}\,\delta\sigma, \tag{4.15}$$

$$\dot{e}_\gamma = -\kappa\,\dot{\gamma}^*\,e_c + \left(a_{\gamma r} + \frac{\dot{\gamma}^*}{r^*}\right) e_h + \left(a_{\gamma\gamma} + \dot{\gamma}^*\tan\gamma^*\right) e_\gamma + \left(a_{\gamma V} - \frac{\dot{\gamma}^*}{V^*}\right) e_V + b_{\gamma\alpha}\,\delta\alpha + b_{\gamma\sigma}\,\delta\sigma, \tag{4.16}$$

$$\dot{e}_V = -\kappa\,\dot{V}^*\,e_c + \left(a_{Vr} + \frac{\dot{V}^*}{r^*}\right) e_h + \left(a_{V\gamma} + \dot{V}^*\tan\gamma^*\right) e_\gamma + \left(a_{VV} - \frac{\dot{V}^*}{V^*}\right) e_V + b_{V\alpha}\,\delta\alpha. \tag{4.17}$$

All a_{ij} and b_{ij} are the Chapter 3 expressions evaluated on the reference at the current s. The heading error does not appear in the $\dot{e}_\gamma$ and $\dot{e}_V$ rows because f_γ and f_V do not depend on ψ once rotation is dropped.

Step 5: state-space form

With $\boldsymbol{e} = [\,e_c,\ e_h,\ e_\psi,\ e_\gamma,\ e_V\,]^T$ and $\delta\boldsymbol{u} = [\,\delta\alpha,\ \delta\sigma\,]^T$:

$$\dot{\boldsymbol{e}} = A_e(s)\,\boldsymbol{e} + B_e(s)\,\delta\boldsymbol{u}, \tag{4.18}$$

$$A_e = \begin{bmatrix} 0 & 0 & \dot{s}^* & 0 & 0 \\ -V^*\kappa\sin\gamma^* & \dfrac{V^*\sin\gamma^*}{r^*} & 0 & \dfrac{V^*}{\cos\gamma^*} & 0 \\ -\kappa\dot{\psi}^* & a_{\psi r} + \dfrac{\dot{\psi}^*}{r^*} & a_{\psi\psi} & a_{\psi\gamma} + \dot{\psi}^*\tan\gamma^* & a_{\psi V} - \dfrac{\dot{\psi}^*}{V^*} \\ -\kappa\dot{\gamma}^* & a_{\gamma r} + \dfrac{\dot{\gamma}^*}{r^*} & 0 & a_{\gamma\gamma} + \dot{\gamma}^*\tan\gamma^* & a_{\gamma V} - \dfrac{\dot{\gamma}^*}{V^*} \\ -\kappa\dot{V}^* & a_{Vr} + \dfrac{\dot{V}^*}{r^*} & 0 & a_{V\gamma} + \dot{V}^*\tan\gamma^* & a_{VV} - \dfrac{\dot{V}^*}{V^*} \end{bmatrix}, \qquad B_e = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ b_{\psi\alpha} & b_{\psi\sigma} \\ b_{\gamma\alpha} & b_{\gamma\sigma} \\ b_{V\alpha} & 0 \end{bmatrix}. \tag{4.19}$$

4.6 Feedforward and why the model has no disturbance term

At $\boldsymbol{e} = 0$ and $\delta\boldsymbol{u} = 0$, every row of the nonlinear error dynamics is exactly zero, because the reference input $\boldsymbol{u}^*(s)$ reproduces the reference motion. In Rajamani's lane-keeping model, the road curvature enters as a disturbance term ($\dot{\psi}_{\text{des}}$) that the feedback must reject unless a feedforward steering angle is added. Here the planner already supplies that feedforward: $\alpha^*(s)$ and $\sigma^*(s)$ are the inputs that fly the path exactly. The commands are

$$\alpha_c = \alpha^*(s) + \delta\alpha, \qquad \sigma_c = \sigma^*(s) + \delta\sigma, \tag{4.20}$$

and the feedback only corrects deviations.

4.7 Arc length as the independent variable

To schedule on s rather than time, divide by $\dot{s}$. To first order, $d\boldsymbol{e}/ds = (1/\dot{s}^*)\,\dot{\boldsymbol{e}}$, because the correction to $\dot{s}$ multiplies $\dot{\boldsymbol{e}}$, which is already first order:

$$\frac{d\boldsymbol{e}}{ds} = \frac{1}{\dot{s}^*}A_e(s)\,\boldsymbol{e} + \frac{1}{\dot{s}^*}B_e(s)\,\delta\boldsymbol{u}. \tag{4.21}$$

4.8 Structure and controllability

- **Lateral chain:** $\delta\sigma \xrightarrow{b_{\psi\sigma}} e_\psi \xrightarrow{\dot{s}^*} e_c$. A double integrator from bank to crossrange, exactly like steering to lateral offset in Rajamani's model. $b_{\psi\sigma} = a_L \cos\sigma^*/(V^* \cos\gamma^*)$ is largest near wings-level and vanishes at $\sigma^* = \pm 90°$.
- **Vertical chain:** $(\delta\alpha, \delta\sigma) \to e_\gamma \to e_h$, with $b_{\gamma\sigma} = -a_L \sin\sigma^*/V^*$. Bank changes vertical lift only when the reference is already banked.
- **Energy:** $\delta\alpha \xrightarrow{b_{V\alpha}} e_V$ through drag, and $e_\gamma \xrightarrow{a_{V\gamma} = -g\cos\gamma} e_V$ through gravity. This is what replaces the throttle loop of a powered vehicle.
- **Locality.** The model is valid for small errors. Large crossrange corrections in entry guidance are made by *bank reversals* (sign changes of σ), which are outside any linear model and are usually handled by logic, such as a heading-error corridor.

4.9 What was neglected

- Earth rotation and J_2 in the partials (Section 3.2). Keep them by using exact Jacobians in code.
- The effect of e_c on latitude, which would add $a_{\psi\phi}\,(-\sin\psi^*/R_0)\,e_c$ to the heading row: about 10^{-7} rad/s per km of crossrange.
- The difference between the heading reference at the vehicle and at the projected point (meridian convergence), of order $e_c \tan\phi/R_0$: about $0.005°$ per km.

Chapter 5

LQR design on the error-state model

5.1 Cost and weights

$$J = \int \left(\boldsymbol{e}^T Q\, \boldsymbol{e} + \delta\boldsymbol{u}^T R\, \delta\boldsymbol{u} \right) dt, \qquad \delta\boldsymbol{u} = -K(s)\, \boldsymbol{e}. \tag{5.1}$$

Bryson's rule sets diagonal weights from the largest acceptable value of each quantity: $Q_{ii} = 1/e_{i,\max}^2$ and $R_{jj} = 1/\delta u_{j,\max}^2$. Reasonable starting values:

Error	Largest acceptable	Q_{ii}	Input	Largest correction	R_{jj}
e_c	1 km	10^{-6} m^{-2}	$\delta\alpha$	2°	$1/(0.0349)^2$
e_h	500 m	4×10^{-6} m^{-2}	$\delta\sigma$	10°	$1/(0.1745)^2$
e_ψ	1°	$1/(0.0175)^2$			
e_γ	0.5°	$1/(0.0087)^2$			
e_V	20 m/s	2.5×10^{-3} (m/s)$^{-2}$			

5.2 Computing the gains

Frozen-point (gain scheduling). At each reference node s_k, solve the algebraic Riccati equation

$$A_e^T P + P A_e - P B_e R^{-1} B_e^T P + Q = 0, \qquad K_k = R^{-1} B_e^T P, \tag{5.2}$$

and interpolate $K(s)$ between nodes (`scipy.linalg.solve_continuous_are`). This is the form that carries over directly to aircraft gain scheduling. It assumes the dynamics change slowly relative to the closed-loop bandwidth.

Finite horizon. The trajectory ends at a target, so the more principled form integrates the differential Riccati equation backward from the end:

$$-\dot{P} = A_e^T P + P A_e - P B_e R^{-1} B_e^T P + Q, \qquad P(t_f) = Q_f, \qquad K(t) = R^{-1} B_e^T P(t). \tag{5.3}$$

In arc length, use $A_e/\dot{s}^*$, $B_e/\dot{s}^*$ and weights $Q/\dot{s}^*$, $R/\dot{s}^*$ (since $dt = ds/\dot{s}$), integrating backward in s.

5.3 The control law

$$\alpha_c = \alpha^*(s) - K_{\alpha c}e_c - K_{\alpha h}e_h - K_{\alpha\psi}e_\psi - K_{\alpha\gamma}e_\gamma - K_{\alpha V}e_V, \tag{5.4}$$
$$\sigma_c = \sigma^*(s) - K_{\sigma c}e_c - K_{\sigma h}e_h - K_{\sigma\psi}e_\psi - K_{\sigma\gamma}e_\gamma - K_{\sigma V}e_V, \tag{5.5}$$

then saturate: $\alpha_{\text{lo}}(M) \le \alpha_c \le \alpha_{\text{hi}}(M)$, $|\sigma_c| \le \sigma_{\max}$, and rate-limit if the inner loop needs it.

5.4 Getting A and B numerically

The hand-derived partials are for understanding and for sanity checks. In code, take exact Jacobians of the full model (rotation, J_2, spline aerodynamics) with CasADi. The planner's model treats α and σ as states; for guidance they are inputs, so split the state:

```
import casadi as ca
x = ca.MX.sym("x", 6)                    # [r, lon, lat, V, gamma, psi]
u = ca.MX.sym("u", 2)                    # [alpha, sigma]: inputs to the guidance loop
xdot = model.f(ca.vertcat(x, u), ca.DM.zeros(2))[:6]
AB = ca.Function("AB", [x, u], [ca.jacobian(xdot, x), ca.jacobian(xdot, u)])
A, B = AB(res.x[:6, k], res.x[6:, k])  # at node k of the plan
```

The entries of A and B correspond to the a_{ij} and b_{ij} of Chapter 3. A_e and B_e then follow from Equation (4.19) using the reference rates $\dot{V}^*$, $\dot{\gamma}^*$, $\dot{\psi}^*$ and the curvature κ. Alternatively, write the nonlinear error dynamics Equation (4.5)–Equation (4.9) as a CasADi function of $(\boldsymbol{e}, \delta\boldsymbol{u})$ and differentiate it at zero.

5.5 Checking the design

Classical verification applies whatever the design method: break the loop at each input ($\delta\alpha$, $\delta\sigma$) of the closed loop, compute the loop transfer function, and report gain and phase margins at frozen points along the path. Continuous-time LQR with full-state feedback guarantees at least 60° of phase margin and a gain margin from -6 dB to $+\infty$ at the plant input, but those guarantees disappear once an estimator, actuator lag or time delay is in the loop (Doyle, 1978), so they must be checked. Then run Monte Carlo dispersions (density, aerodynamic coefficients, mass, initial state) in the simulator.

Chapter 6

Inner-loop attitude control: classical and NDI

In a 6-DOF simulation, guidance commands α_c, σ_c and $\beta_c = 0$, and an attitude control loop makes the airframe follow them using control surfaces (and reaction control jets at low dynamic pressure). This chapter gives the models for that loop and the structure of the two common designs. It uses *forces* L, D, Y (newtons) rather than the accelerations of earlier chapters, with $a_L = L/m$.

6.1 Notation for this chapter

Symbol	Meaning
p, q, r	body-axis roll, pitch, yaw rates
α, β	angle of attack, sideslip
μ	wind-axis bank angle (the σ of earlier chapters)
χ	flight-path heading (the ψ of earlier chapters)
L, D, Y	lift, drag, side force in wind axes [N]
$\mathcal{L}, \mathcal{M}, \mathcal{N}$	rolling, pitching, yawing moments about the center of mass [N m]
$\boldsymbol{\delta}$	vector of control-surface deflections

For the fast attitude dynamics, the Earth is treated as flat and non-rotating. The neglected terms are of order V/r and Ω (about 10^{-3} rad/s), far below attitude bandwidths (about 1 rad/s).

6.2 Rotational dynamics

Euler's equation $I\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times I\boldsymbol{\omega} = \boldsymbol{M}$, with the symmetric inertia tensor $I = \begin{bmatrix} I_{xx} & 0 & -I_{xz} \\ 0 & I_{yy} & 0 \\ -I_{xz} & 0 & I_{zz} \end{bmatrix}$, written out by component:

$$I_{xx}\dot{p} - I_{xz}\dot{r} = \mathcal{L} + (I_{yy} - I_{zz})\, qr + I_{xz}\, pq, \tag{6.1}$$

$$I_{yy}\dot{q} = \mathcal{M} + (I_{zz} - I_{xx})\, pr + I_{xz}\,(r^2 - p^2), \tag{6.2}$$

$$I_{zz}\dot{r} - I_{xz}\dot{p} = \mathcal{N} + (I_{xx} - I_{yy})\, pq - I_{xz}\, qr. \tag{6.3}$$

The HL-20 data set $I_{xz} = 0$, so each axis can be solved on its own:

$$\dot{p} = \frac{1}{I_{xx}} \left[\mathcal{L} + (I_{yy} - I_{zz})\, qr \right], \tag{6.4}$$

$$\dot{q} = \frac{1}{I_{yy}} \left[\mathcal{M} + (I_{zz} - I_{xx})\, pr \right], \tag{6.5}$$

$$\dot{r} = \frac{1}{I_{zz}} \left[\mathcal{N} + (I_{xx} - I_{yy})\, pq \right]. \tag{6.6}$$

6.3 Aerodynamic moments

$$\mathcal{L} = \bar{q} S b \left[C_{l_\beta} \beta + C_{l_p} \frac{pb}{2V} + C_{l_r} \frac{rb}{2V} + C_{l,\boldsymbol{\delta}}\, \boldsymbol{\delta} \right], \tag{6.7}$$

$$\mathcal{M} = \bar{q} S \bar{c} \left[C_{m_0}(\alpha, M) + C_{m_q} \frac{q\bar{c}}{2V} + C_{m,\boldsymbol{\delta}}\, \boldsymbol{\delta} \right], \tag{6.8}$$

$$\mathcal{N} = \bar{q} S b \left[C_{n_\beta} \beta + C_{n_p} \frac{pb}{2V} + C_{n_r} \frac{rb}{2V} + C_{n,\boldsymbol{\delta}}\, \boldsymbol{\delta} \right]. \tag{6.9}$$

The `hgv3` aero file contains C_m, the sideslip derivatives C_{Y_β}, C_{l_β}, C_{n_β} (per *degree* of β) and the damping derivatives C_{m_q}, C_{l_p}, C_{l_r}, C_{n_p}, C_{n_r}. The control-surface terms $C_{\cdot,\boldsymbol{\delta}}$ are not in that file; take them from `hgv6`'s effector model.

Throughout, split each moment into a part independent of the controls and a part linear in them:

$$\mathcal{L} = \mathcal{L}_0(\boldsymbol{x}) + \bar{q} S b\, C_{l,\boldsymbol{\delta}}\, \boldsymbol{\delta}, \quad \mathcal{M} = \mathcal{M}_0(\boldsymbol{x}) + \bar{q} S \bar{c}\, C_{m,\boldsymbol{\delta}}\, \boldsymbol{\delta}, \quad \mathcal{N} = \mathcal{N}_0(\boldsymbol{x}) + \bar{q} S b\, C_{n,\boldsymbol{\delta}}\, \boldsymbol{\delta}. \tag{6.10}$$

6.4 Wind-axis attitude kinematics

These relate the body rates to the rates of the controlled angles. They are the key equations for both designs:

$$\dot{\alpha} = q - \tan\beta \left(p \cos\alpha + r \sin\alpha \right) - \frac{L - mg \cos\gamma \cos\mu}{mV \cos\beta}, \tag{6.11}$$

$$\dot{\beta} = p \sin\alpha - r \cos\alpha + \frac{Y + mg \cos\gamma \sin\mu}{mV}, \tag{6.12}$$

$$\dot{\mu} = \frac{p \cos\alpha + r \sin\alpha}{\cos\beta} + \frac{L \left(\tan\gamma \sin\mu + \tan\beta \right) + Y \tan\gamma \cos\mu - mg \cos\gamma \cos\mu \tan\beta}{mV}. \tag{6.13}$$

The flight-path equations in the same frame are

$$\dot{\gamma} = \frac{L \cos\mu - Y \sin\mu - mg \cos\gamma}{mV}, \qquad \dot{\chi} = \frac{L \sin\mu + Y \cos\mu}{mV \cos\gamma}. \tag{6.14}$$

With $Y = 0$ and the curved-Earth terms restored, these are the 3-DOF equations Equations (3.7) and (3.8) with $\mu = \sigma$ and $\chi = \psi$.

Roll about the velocity vector. In Equation (6.13), the body rates enter as $p \cos\alpha + r \sin\alpha$, the roll rate about the velocity vector. At the high angles of attack used in entry (20° to 40°), rolling about the body x axis alone ($r = 0$) would, from Equation (6.12), convert directly into sideslip at the rate $p \sin\alpha$. Banking must therefore be done about the velocity vector, with $r = p \tan\alpha$. Both designs below have to respect this.

6.5 Classical gain-scheduled design

6.5.1 Linearization about a wings-level trim

At each design point $(M, \bar{q}, \alpha^*)$ along the trajectory, linearize about $\beta = 0$, $\mu = 0$, $p = r = 0$. The trim values of V and γ are held fixed (they change slowly compared with attitude).

Longitudinal (short period). From Equations (6.5) and (6.11):

$$\delta\dot{\alpha} = \delta q - \frac{\delta L}{mV} = Z_\alpha\,\delta\alpha + \delta q + Z_\delta\,\delta\delta_e, \tag{6.15}$$

$$\delta\dot{q} = M_\alpha\,\delta\alpha + M_q\,\delta q + M_\delta\,\delta\delta_e, \tag{6.16}$$

with the dimensional derivatives

$$Z_\alpha = -\frac{\bar{q}SC_{L_\alpha}}{mV}, \qquad Z_\delta = -\frac{\bar{q}SC_{L,\delta_e}}{mV},$$
$$M_\alpha = \frac{\bar{q}S\bar{c}\,C_{m_\alpha}}{I_{yy}}, \qquad M_q = \frac{\bar{q}S\bar{c}^2C_{m_q}}{2VI_{yy}}, \qquad M_\delta = \frac{\bar{q}S\bar{c}\,C_{m,\delta_e}}{I_{yy}}. \tag{6.17}$$

Here δ_e is whichever symmetric surface produces pitching moment. The characteristic equation is

$$s^2 - (Z_\alpha + M_q)\,s + (Z_\alpha M_q - M_\alpha) = 0, \qquad \omega_{sp}^2 = Z_\alpha M_q - M_\alpha, \quad 2\zeta_{sp}\omega_{sp} = -(Z_\alpha + M_q)\,. \tag{6.18}$$

Lateral-directional. From Equations (6.4), (6.6), (6.12) and (6.13):

$$\delta\dot{\beta} = Y_\beta\,\delta\beta + \sin\alpha^*\,\delta p - \cos\alpha^*\,\delta r + \frac{g\cos\gamma^*}{V}\,\delta\mu + Y_{\boldsymbol{\delta}}\,\delta\boldsymbol{\delta}, \tag{6.19}$$

$$\delta\dot{p} = L_\beta\,\delta\beta + L_p\,\delta p + L_r\,\delta r + L_{\boldsymbol{\delta}}\,\delta\boldsymbol{\delta}, \tag{6.20}$$

$$\delta\dot{r} = N_\beta\,\delta\beta + N_p\,\delta p + N_r\,\delta r + N_{\boldsymbol{\delta}}\,\delta\boldsymbol{\delta}, \tag{6.21}$$

$$\delta\dot{\mu} = \left(\frac{L^* - mg\cos\gamma^*}{mV} + Y_\beta\tan\gamma^*\right)\delta\beta + \cos\alpha^*\,\delta p + \sin\alpha^*\,\delta r + \frac{L^*\tan\gamma^*}{mV}\,\delta\mu + \tan\gamma^*\,Y_{\boldsymbol{\delta}}\,\delta\boldsymbol{\delta}, \tag{6.22}$$

with

$$Y_\beta = \frac{\bar{q}SC_{Y_\beta}}{mV}, \qquad L_\beta = \frac{\bar{q}Sb\,C_{l_\beta}}{I_{xx}}, \qquad L_p = \frac{\bar{q}Sb^2C_{l_p}}{2VI_{xx}}, \qquad L_r = \frac{\bar{q}Sb^2C_{l_r}}{2VI_{xx}},$$
$$N_\beta = \frac{\bar{q}Sb\,C_{n_\beta}}{I_{zz}}, \qquad N_p = \frac{\bar{q}Sb^2C_{n_p}}{2VI_{zz}}, \qquad N_r = \frac{\bar{q}Sb^2C_{n_r}}{2VI_{zz}}. \tag{6.23}$$

Convert the `hgv3` sideslip derivatives from per degree to per radian (multiply by $180/\pi$) before using them here.

6.5.2 Control structure

- **Pitch:** an inner pitch-rate loop for damping, and an outer proportional-integral (PI) loop on α that commands it: $q_c = K_\alpha(\alpha_c - \alpha) + K_{I\alpha}\int(\alpha_c - \alpha)\,dt$, then $\delta_e = K_q(q_c - q)$.
- **Roll:** a PI loop on μ commanding the stability-axis roll rate $p_{s,c} = p_c\cos\alpha + r_c\sin\alpha$, implemented as p_c and $r_c = p_c\tan\alpha$ so the roll is about the velocity vector.

- **Yaw:** drive β to zero (turn coordination), with yaw-rate damping through a washout filter so steady turning is not opposed.
- **Scheduling:** control power scales with $\bar{q}$, so gains are scheduled on $\bar{q}$ and Mach, at minimum $K \propto 1/\bar{q}$ to hold loop gain constant.
- **Requirements:** at every design point, with the loop broken at each actuator, at least 6 dB of gain margin and 45° of phase margin. Keep bandwidths separated: guidance about 0.05–0.2 rad/s, attitude about 1–3 rad/s, actuators 20 rad/s or more.

6.6 Nonlinear dynamic inversion (NDI)

6.6.1 The idea, written out

Suppose a set of outputs $\boldsymbol{y}$ obeys

$$\dot{\boldsymbol{y}} = \boldsymbol{f}(\boldsymbol{x}) + G(\boldsymbol{x})\,\boldsymbol{v}, \tag{6.24}$$

where $\boldsymbol{v}$ is the input, $\boldsymbol{f}$ collects everything the input does not affect, and G is square and invertible. Choose the input

$$\boldsymbol{v} = G(\boldsymbol{x})^{-1}\left[\boldsymbol{\nu} - \boldsymbol{f}(\boldsymbol{x})\right]. \tag{6.25}$$

Substituting back into Equation (6.24):

$$\dot{\boldsymbol{y}} = \boldsymbol{f}(\boldsymbol{x}) + G(\boldsymbol{x})\,G(\boldsymbol{x})^{-1}\left[\boldsymbol{\nu} - \boldsymbol{f}(\boldsymbol{x})\right] = \boldsymbol{\nu}. \tag{6.26}$$

The nonlinear system now behaves as a set of decoupled integrators driven by the *pseudo-control* $\boldsymbol{\nu}$. Choose $\boldsymbol{\nu}$ as a simple linear law, for example

$$\nu_i = \dot{y}_{i,c} + K_{P,i}\,(y_{i,c} - y_i) + K_{I,i}\int (y_{i,c} - y_i)\,dt, \tag{6.27}$$

which gives each channel first-order (or, with the integral, second-order) linear error dynamics set by the gains. Conditions: G must be invertible along the trajectory, and the *internal dynamics*, the states not directly controlled (V, γ, χ here), must be stable or handled by another loop (guidance).

For the HL-20, the controlled angles (α, β, μ) depend on the body rates, and the body rates depend on the control surfaces. So the inversion is done in two stages, using time-scale separation: the rates respond much faster than the angles.

6.6.2 Slow loop: angles to body rates

From Equations (6.11) to (6.13), the angle rates are affine in (p, q, r):

$$\begin{bmatrix}\dot{\alpha}\\ \dot{\beta}\\ \dot{\mu}\end{bmatrix} = \begin{bmatrix}f_\alpha\\ f_\beta\\ f_\mu\end{bmatrix} + G_s(\alpha,\beta)\begin{bmatrix}p\\ q\\ r\end{bmatrix}, \tag{6.28}$$

$$f_\alpha = -\frac{L - mg\cos\gamma\cos\mu}{mV\cos\beta}, \tag{6.29}$$

$$f_\beta = \frac{Y + mg\cos\gamma\sin\mu}{mV}, \tag{6.30}$$

$$f_\mu = \frac{L\,(\tan\gamma\sin\mu + \tan\beta) + Y\tan\gamma\cos\mu - mg\cos\gamma\cos\mu\tan\beta}{mV}, \tag{6.31}$$

$$G_s = \begin{bmatrix} -\tan\beta\cos\alpha & 1 & -\tan\beta\sin\alpha \\ \sin\alpha & 0 & -\cos\alpha \\ \dfrac{\cos\alpha}{\cos\beta} & 0 & \dfrac{\sin\alpha}{\cos\beta} \end{bmatrix}, \qquad \det G_s = -\frac{1}{\cos\beta}. \tag{6.32}$$

G_s is invertible for all $|\beta| < 90°$, with

$$G_s^{-1} = \begin{bmatrix} 0 & \sin\alpha & \cos\alpha\cos\beta \\ 1 & 0 & \sin\beta \\ 0 & -\cos\alpha & \sin\alpha\cos\beta \end{bmatrix}. \tag{6.33}$$

With desired angle rates ν_α, ν_β, ν_μ, the commanded body rates are, written out:

$$p_c = \sin\alpha\,(\nu_\beta - f_\beta) + \cos\alpha\cos\beta\,(\nu_\mu - f_\mu)\,, \tag{6.34}$$
$$q_c = (\nu_\alpha - f_\alpha) + \sin\beta\,(\nu_\mu - f_\mu)\,, \tag{6.35}$$
$$r_c = -\cos\alpha\,(\nu_\beta - f_\beta) + \sin\alpha\cos\beta\,(\nu_\mu - f_\mu)\,. \tag{6.36}$$

With $\beta \approx 0$, a pure bank command gives $p_c = \cos\alpha\,\nu'_\mu$ and $r_c = \sin\alpha\,\nu'_\mu$, so $r_c/p_c = \tan\alpha$: the inversion automatically rolls about the velocity vector. The desired rates are simple linear laws:

$$\nu_\alpha = \omega_\alpha(\alpha_c - \alpha), \qquad \nu_\beta = \omega_\beta(0 - \beta), \qquad \nu_\mu = \omega_\mu(\mu_c - \mu), \tag{6.37}$$

optionally with integral terms. L and Y in f_α, f_β, f_μ come from the aerodynamic model at the current state.

6.6.3 Fast loop: body rates to control moments

From Equations (6.4) to (6.6) with the split Equation (6.10):

$$\dot p = \frac{(I_{yy} - I_{zz})\,qr + \mathcal{L}_0}{I_{xx}} + \frac{\bar q S b}{I_{xx}} C_{l,\boldsymbol\delta}\,\boldsymbol\delta, \tag{6.38}$$
$$\dot q = \frac{(I_{zz} - I_{xx})\,pr + \mathcal{M}_0}{I_{yy}} + \frac{\bar q S \bar c}{I_{yy}} C_{m,\boldsymbol\delta}\,\boldsymbol\delta, \tag{6.39}$$
$$\dot r = \frac{(I_{xx} - I_{yy})\,pq + \mathcal{N}_0}{I_{zz}} + \frac{\bar q S b}{I_{zz}} C_{n,\boldsymbol\delta}\,\boldsymbol\delta. \tag{6.40}$$

With desired rate derivatives $\nu_p = \omega_p(p_c - p)$, $\nu_q = \omega_q(q_c - q)$, $\nu_r = \omega_r(r_c - r)$, solve each axis for the moment the surfaces must produce:

$$\mathcal{L}_c = I_{xx}\,\nu_p - (I_{yy} - I_{zz})\,qr - \mathcal{L}_0, \tag{6.41}$$
$$\mathcal{M}_c = I_{yy}\,\nu_q - (I_{zz} - I_{xx})\,pr - \mathcal{M}_0, \tag{6.42}$$
$$\mathcal{N}_c = I_{zz}\,\nu_r - (I_{xx} - I_{yy})\,pq - \mathcal{N}_0. \tag{6.43}$$

Control allocation. The surfaces must satisfy

$$\bar q S \begin{bmatrix} b\,C_{l,\boldsymbol\delta} \\ \bar c\,C_{m,\boldsymbol\delta} \\ b\,C_{n,\boldsymbol\delta} \end{bmatrix} \boldsymbol\delta \equiv B_\delta\,\boldsymbol\delta = \begin{bmatrix} \mathcal{L}_c \\ \mathcal{M}_c \\ \mathcal{N}_c \end{bmatrix}. \tag{6.44}$$

With exactly three independent effectors, $\boldsymbol\delta = B_\delta^{-1}[\mathcal{L}_c, \mathcal{M}_c, \mathcal{N}_c]^T$. With more than three, use a weighted pseudo-inverse, $\boldsymbol\delta = W^{-1}B_\delta^T\left(B_\delta W^{-1} B_\delta^T\right)^{-1}[\mathcal{L}_c, \mathcal{M}_c, \mathcal{N}_c]^T$, then clip to deflection limits (or use a constrained allocator).

6.6.4 Incremental NDI (INDI)

NDI needs accurate $\mathcal{L}_0$, $\mathcal{M}_0$, $\mathcal{N}_0$. INDI avoids most of that model. Write the rate dynamics as $\dot{\boldsymbol{\omega}} = \boldsymbol{f}_\omega(\boldsymbol{x}) + G_\omega \boldsymbol{\delta}$ with $G_\omega = \mathrm{diag}(I_{xx}, I_{yy}, I_{zz})^{-1} B_\delta$. At the previous sample, the measured angular acceleration is $\dot{\boldsymbol{\omega}}_0 = \boldsymbol{f}_\omega(\boldsymbol{x}_0) + G_\omega \boldsymbol{\delta}_0$. Subtracting, and assuming $\boldsymbol{f}_\omega$ changes little over one sample:

$$\dot{\boldsymbol{\omega}} - \dot{\boldsymbol{\omega}}_0 \approx G_\omega \left(\boldsymbol{\delta} - \boldsymbol{\delta}_0\right) \quad \Rightarrow \quad \boldsymbol{\delta} = \boldsymbol{\delta}_0 + G_\omega^{-1} \left(\boldsymbol{\nu}_\omega - \dot{\boldsymbol{\omega}}_0\right). \tag{6.45}$$

Only the control effectiveness G_ω is needed. The costs are an angular acceleration estimate (filtered differentiation of the rate gyros), knowledge of the actual actuator positions $\boldsymbol{\delta}_0$, and matched filtering of the two.

6.6.5 Practical points

- **Bandwidth separation:** $\omega_p, \omega_q, \omega_r$ about 3–5 times $\omega_\alpha, \omega_\beta, \omega_\mu$, which are about 3–5 times the guidance bandwidth. Actuators faster still.
- **Saturation:** when surfaces saturate the inversion is no longer exact; use anti-windup on integrators and consider pseudo-control hedging.
- **Verification:** NDI's nominal loop is an integrator (ν/s), which looks perfectly robust. Real margins come from actuator dynamics, delays and model error, so linearize the closed loop with those included, break it at the actuators, and check margins exactly as for the classical design. Repeat with perturbed control effectiveness and moment coefficients.

6.7 Comparison

	Classical, gain-scheduled	NDI	INDI
Model needed	linear models at design points	full moment model and effectiveness	effectiveness only
Scheduling	explicit gain tables	built in	built in
Sensitivity to aero error	moderate (margins)	high	low
Extra sensors	none	none	angular acceleration, actuator positions
Certification history	long (Shuttle, most aircraft)	established (X-38, modern fighters)	growing (research, eVTOL)

Appendix A

Verification record

Item	How it was checked
Local basis derivatives, $\boldsymbol{\omega}_{LE}$, kinematics, transport terms (Chapter 2)	Symbolic (sympy): every intermediate expression reduces to zero difference.
Force projections and assembled equations Equation (2.39)–Equation (2.44)	Symbolic: derived from Equation (2.5) and compared with the expressions in `hl20_model.py`: identical.
J_2 gravity Equation (2.32)	Numerical against `hgv3.gravity_vector` at 50 random points: relative error below 10^{-15}.
Model against simulator	Open-loop (α, σ) from Mach 15 for 800 s at three bank settings: altitude within 10 m, speed within 0.1 m/s, along-track drift 0.8 m/s.
Partial derivatives (Chapter 3)	All 34 hand expressions against symbolic differentiation with generic C_L, C_D, ρ, a_s: relative error below 10^{-14}; all entries listed as zero confirmed zero.
Simplification error (Section 3.2)	CasADi Jacobians of the full and simplified model along a planned trajectory.
Error-state model Equation (4.19)	Symbolic Jacobian of the nonlinear error dynamics with generic f_V, f_γ, f_ψ: matches entry by entry; error rates vanish at zero error.
CasADi snippet (Section 5.4)	Run against `hl20_model.py` on a planned trajectory.
Euler equations with I_{xz}	Symbolic: identical to $I\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times I\boldsymbol{\omega} = \boldsymbol{M}$.
Wind-axis kinematics Equation (6.11)–Equation (6.13) and $\dot{\gamma}$, $\dot{\chi}$	Numerical against a rigid-body model (finite difference of angles extracted from the attitude matrix and velocity) at 200 random states with sideslip and side force: agreement to 10^{-10}.
G_s^{-1} and the wings-level linearizations	Symbolic.